

# Smart Irrigation System using Cisco Packet Tracer

---

Dilip Seshang TS (23BCT0121)  
VIT Vellore BTech CSE (IoT)  
dilipseshang.ts2023@vitstudent.ac.in

## 1. Aim

The aim of this project is to design and simulate an Internet of Things (IoT)-based Smart Irrigation System using Cisco Packet Tracer. The system is designed to monitor environmental conditions, such as humidity and water level, and automatically control irrigation through lawn sprinklers whenever irrigation is required. The project demonstrates how IoT devices, networking infrastructure, and cloud connectivity can be integrated to improve water management, reduce manual intervention, and promote efficient irrigation in agricultural fields.

## 2. Problem Statement

Agriculture depends heavily on effective water management. Conventional irrigation methods generally rely on manual operation or fixed irrigation schedules, which often result in excessive water consumption, uneven irrigation, and unnecessary labor. These methods fail to consider the real-time environmental conditions of the agricultural field.

To overcome these limitations, an IoT-enabled Smart Irrigation System is proposed. The system continuously monitors humidity and water level using IoT devices deployed in the field. Based on predefined automation rules, the system automatically activates or deactivates lawn sprinklers without requiring human intervention. The solution also enables remote monitoring through a laptop connected to the IoT network. This approach improves irrigation efficiency, conserves water resources, and supports sustainable agricultural practices.

## 3. Scope of the Solution

The Smart Irrigation System developed in this project focuses on automating irrigation using IoT technology and networking principles. The solution provides a centralized platform for monitoring environmental conditions and controlling irrigation devices.

The scope of this project includes:

- Continuous monitoring of humidity using the Humiture Monitor.
- Continuous monitoring of water level using the Water Level Monitor.
- Automatic operation of lawn sprinklers based on predefined automation rules.
- Wireless communication between IoT devices and the Home Gateway.
- Real-time monitoring of sensor values and sprinkler status through a laptop.
- Cloud-enabled communication using networking devices available in Cisco Packet Tracer.
- Demonstration of an IoT-based smart agriculture application.

### **Future Scope**

The system can be further improved by integrating:

- Soil moisture sensors for more accurate irrigation decisions.
- Weather forecasting APIs to avoid irrigation during rainfall.
- Mobile application support for remote monitoring and control.
- Cloud-based analytics for long-term irrigation analysis.
- Artificial Intelligence and Machine Learning for predictive irrigation scheduling.
- SMS and Email alerts to notify farmers about field conditions.

## **4. Overview / Architecture of the Solution**

The Smart Irrigation System consists of two major layers: the **Cloud Computing Layer** and the **Edge Computing Layer**.

The Cloud Computing Layer contains the networking infrastructure required to provide Internet connectivity and IoT communication. It consists of the Cloud, Cisco Router, Cisco Switch, DNS Server, IoT Server, Cable Modem, Cell Tower, Central Office Server, and Smartphone.

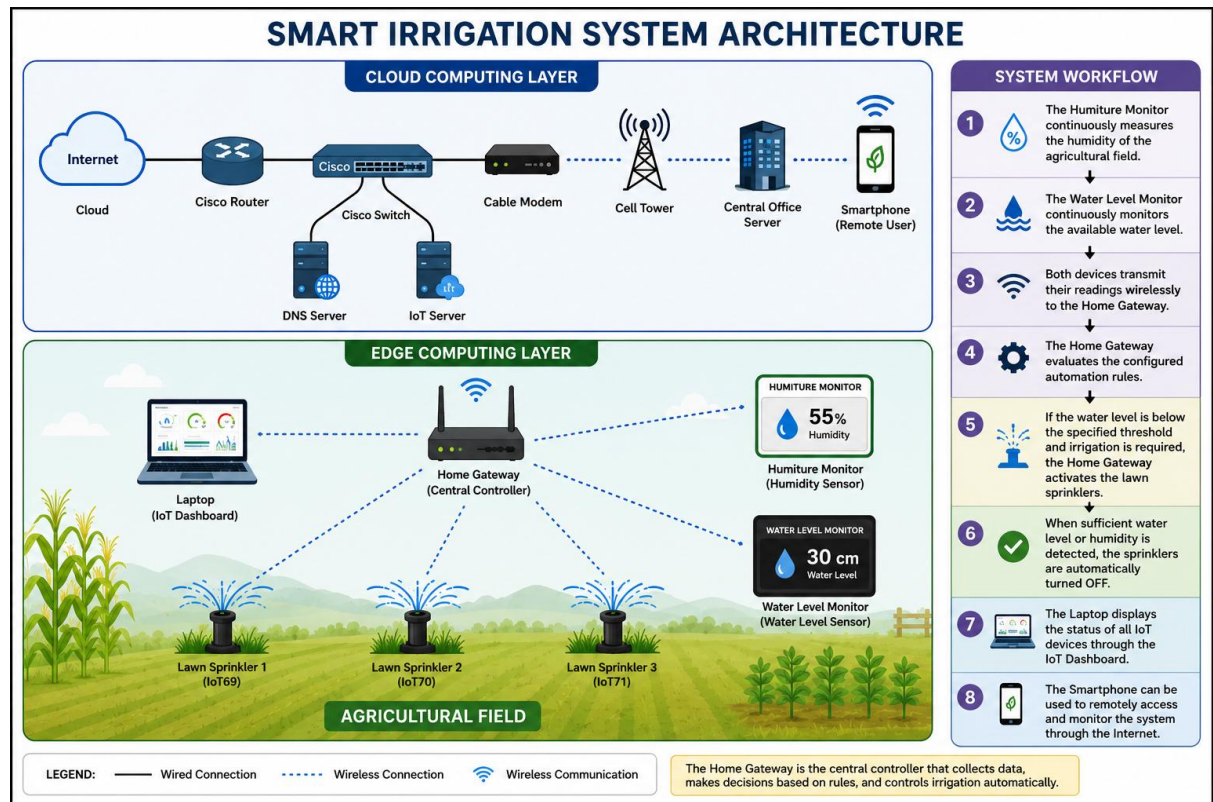
The Edge Computing Layer represents the agricultural field where all IoT devices are deployed. It consists of the Home Gateway, Laptop, Humiture Monitor, Water Level Monitor, and three Lawn Sprinklers.

The Home Gateway acts as the central controller of the system. It receives environmental data from the Humiture Monitor and Water Level Monitor through wireless communication. The Home Gateway evaluates predefined automation rules and automatically controls the lawn sprinklers whenever irrigation is required. The laptop provides an interface to monitor sensor readings, view device status, and configure automation rules.

### **System Workflow**

1. The Humiture Monitor continuously measures the humidity of the agricultural field.
2. The Water Level Monitor continuously monitors the available water level.
3. Both devices transmit their readings wirelessly to the Home Gateway.

4. The Home Gateway evaluates the configured automation rules.
5. If the water level is below the specified threshold and irrigation is required, the Home Gateway activates the lawn sprinklers.
6. When sufficient water level or humidity is detected, the sprinklers are automatically turned OFF.
7. The Laptop displays the status of all IoT devices through the IoT Dashboard.
8. The Smartphone can be used to remotely access and monitor the system through the Internet.



## 5. Required Components to Develop the Solution

The following hardware and software resources were used to develop the Smart Irrigation System.

### Networking Components

| Component         | Quantity | Purpose                                     |
|-------------------|----------|---------------------------------------------|
| Cloud-PT          | 1        | Simulates Internet connectivity             |
| Cisco 2811 Router | 1        | Routes data between cloud and local network |

| Component             | Quantity | Purpose                                            |
|-----------------------|----------|----------------------------------------------------|
| Cisco 2960 Switch     | 1        | Connects networking devices within the LAN         |
| Cable Modem           | 1        | Connects Home Gateway to the Internet              |
| Home Gateway (DLC100) | 1        | Central IoT controller                             |
| DNS Server            | 1        | Resolves domain names into IP addresses            |
| IoT Server            | 1        | Registers and manages IoT devices                  |
| Cell Tower            | 1        | Provides wireless communication for the smartphone |
| Central Office Server | 1        | Simulates the Internet Service Provider            |
| Smartphone            | 1        | Enables remote monitoring                          |
| Laptop                | 1        | Used for monitoring and configuring IoT devices    |

### **IoT Devices**

| Device              | Quantity | Function                          |
|---------------------|----------|-----------------------------------|
| Humiture Monitor    | 1        | Monitors environmental humidity   |
| Water Level Monitor | 1        | Monitors water level              |
| Lawn Sprinkler      | 3        | Performs irrigation automatically |

### **Communication Media**

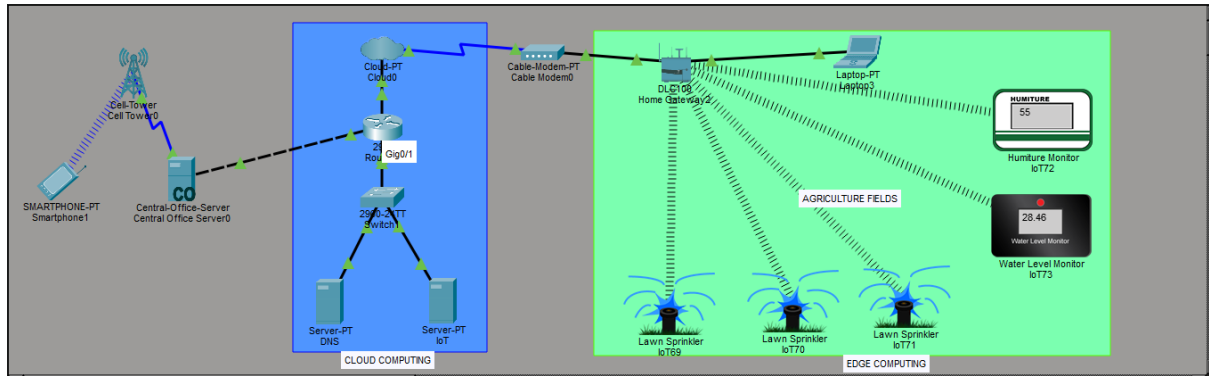
- Copper Straight Through Cable
- Coaxial Cable
- Wireless IoT Communication

### **Software Used**

- Cisco Packet Tracer 9.0.0
- GitHub (Project Repository)

## **6. Simulated Circuit (Logical System View)**

The Smart Irrigation System was simulated using Cisco Packet Tracer 9.0.0.



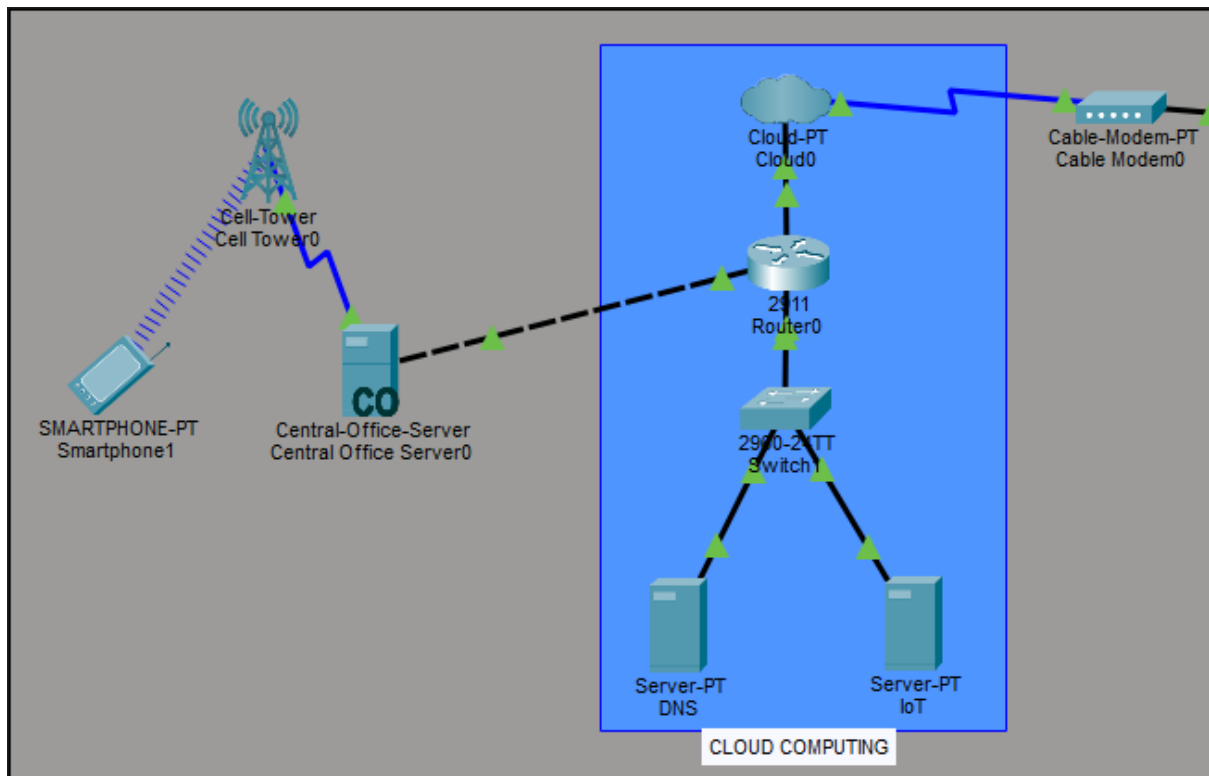
The logical topology consists of two interconnected sections:

### Cloud Computing Layer

The cloud computing layer provides Internet connectivity and communication services.

Components included:

- Cloud
- Cisco Router
- Cisco Switch
- DNS Server
- IoT Server
- Cable Modem
- Cell Tower
- Central Office Server

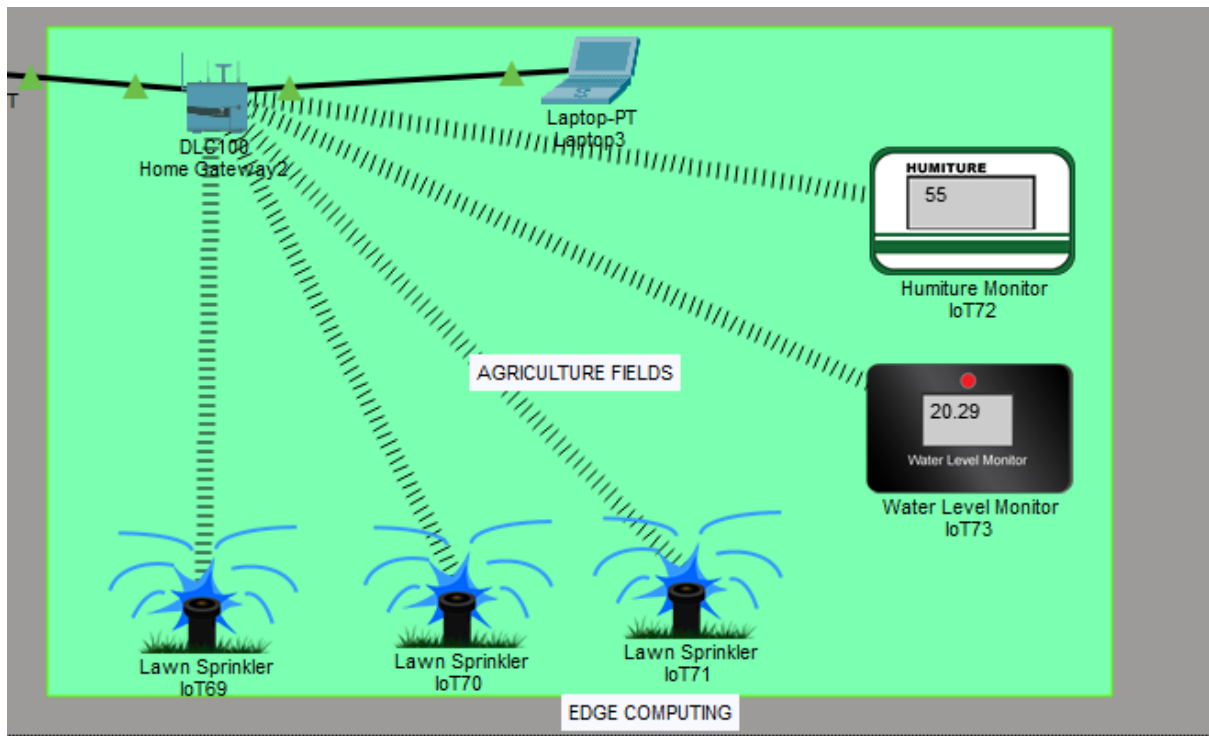


## Edge Computing Layer

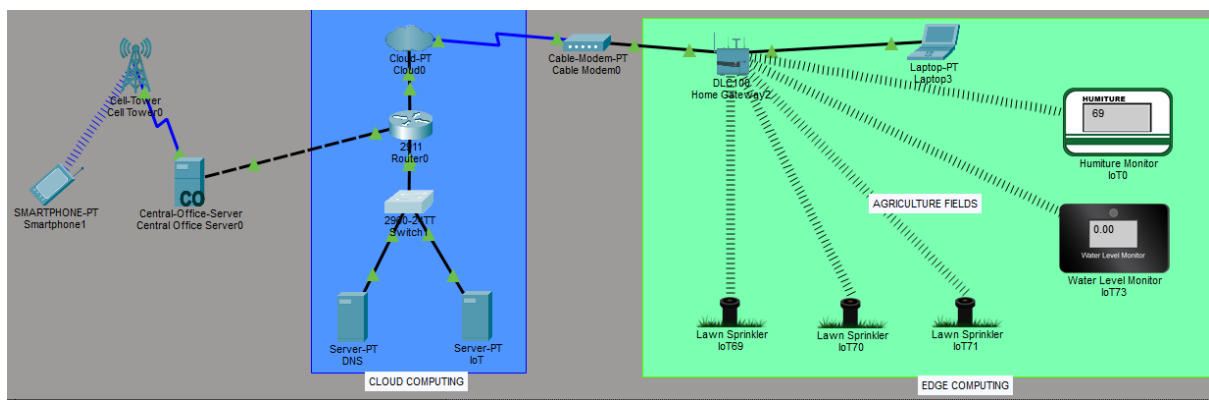
The edge computing layer represents the agricultural field.

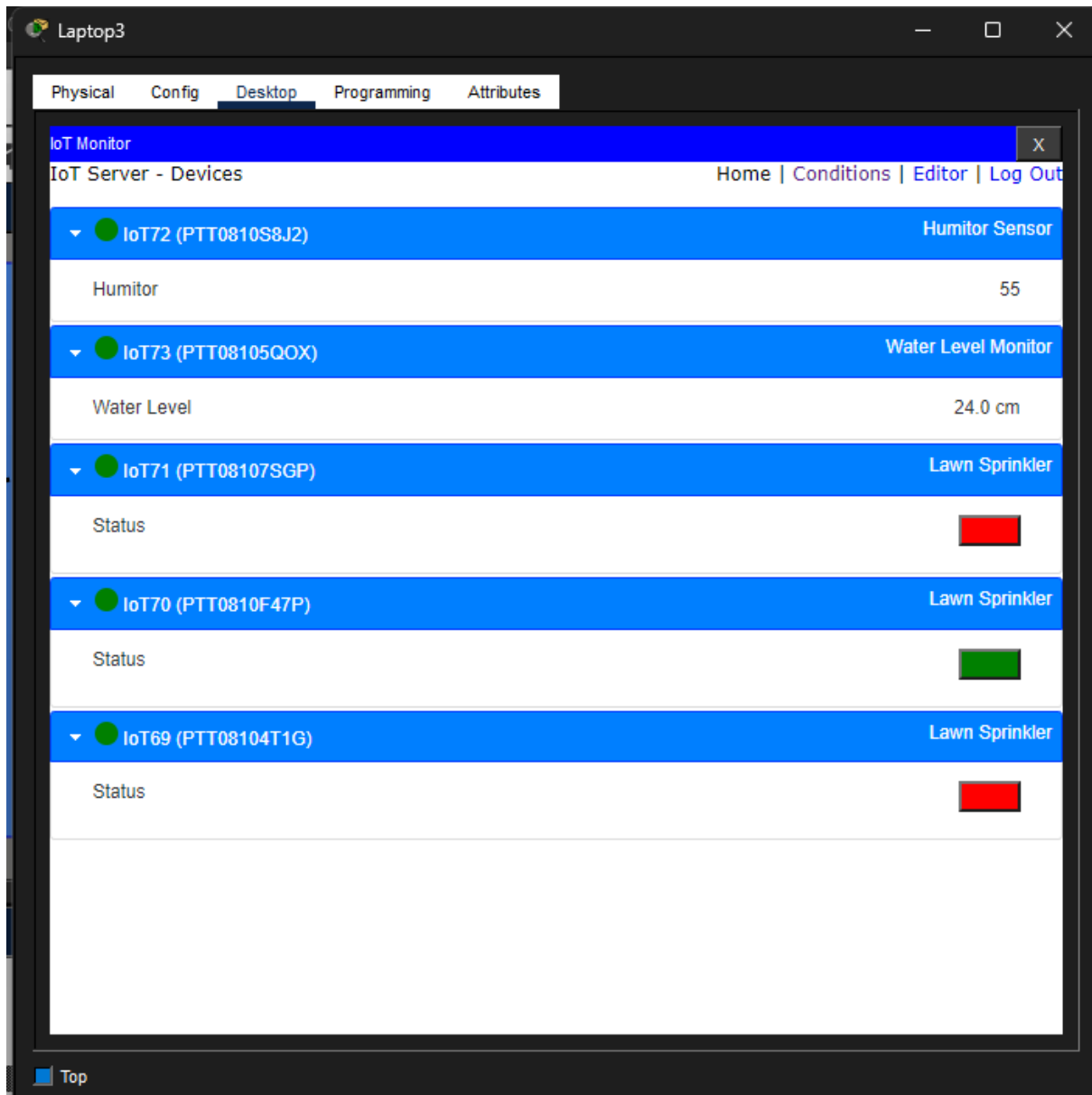
Components included:

- Home Gateway
- Laptop
- Humiture Monitor
- Water Level Monitor
- Lawn Sprinkler 1
- Lawn Sprinkler 2
- Lawn Sprinkler 3



The Home Gateway wirelessly connects all IoT devices and acts as the central controller. It processes sensor readings received from the Humiture Monitor and Water Level Monitor and automatically controls the lawn sprinklers according to the configured automation rules.





## 7. Execution Video of the Demo

The execution video demonstrates the complete working of the Smart Irrigation System developed in Cisco Packet Tracer.

DRIVE LINK:

[https://drive.google.com/file/d/11ne9G1HlVhT6Vo\\_sETl1Xx1VQ44ECuwm/view?usp=sharing](https://drive.google.com/file/d/11ne9G1HlVhT6Vo_sETl1Xx1VQ44ECuwm/view?usp=sharing)